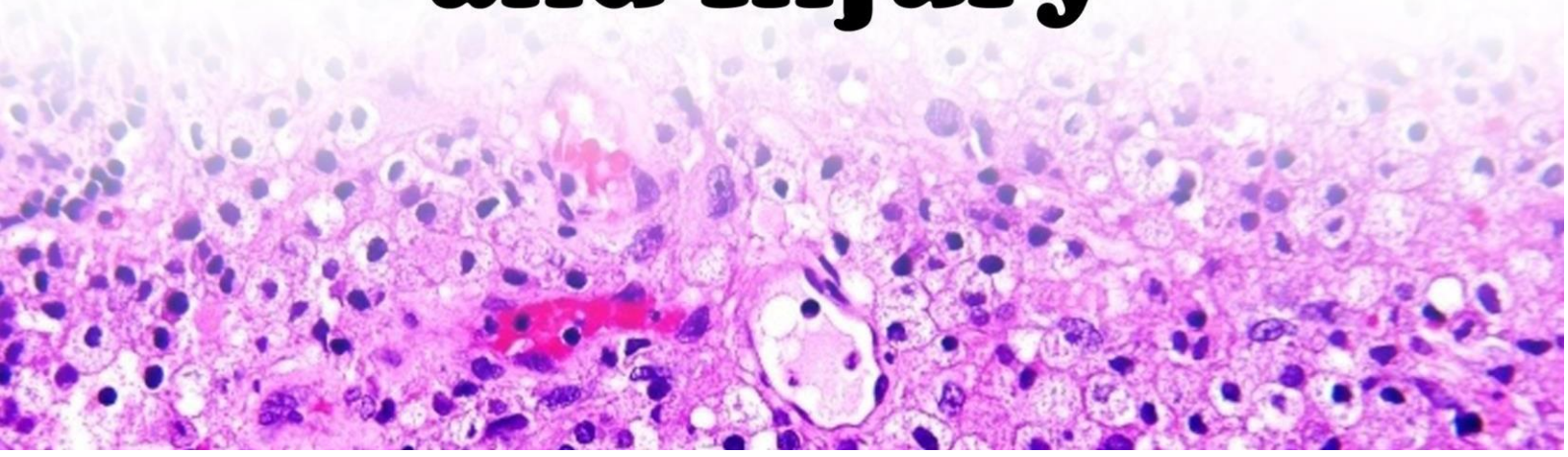
Student Notes

PATHOLOGY

1st Edition

Unit 1

Cellular Adaptation and Injury



Preface

Agam is a group of budding medicos, who are currently doing their under graduation in various Medical Colleges across Tamil Nadu and Pondicherry. The group was initiated on 18th November 2017, in the vision of uniting medicos for various social and professional causes.

We feel delighted to present you Agam Pathology notes prepared by Agam Divide and Rule 2020 Team to guide our fellow medicos to prepare for university examinations.

This is a reference work of 2017 batch medical students from various colleges. The team took effort to refer many books and make them into simple notes. We are not the authors of the following work. The images used in the documents are not copyrighted by us and is obtained from various sources.

Dear readers, we request you to use this material as a reference note, or revision note, or recall notes. Please do not learn the topics for the 1st time from this material, as this contain just the required points, for revision.

Acknowledgement

On behalf of the team, Agam would like to thank all the doctors who taught us Pathology. Agam would like to whole heartedly appreciate and thank everyone who contributed towards the making of this material. A special thanks to Vignesh M, who took the responsibility of leading the team. The following are the name list of the team who worked together, to bring out the material in good form.

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CELLULAR ADAPTATION AND CELL INJURY

ESSAY

1. Necrosis
2. Apoptosis
3. Types of degeneration. Add a note on fat stains
4. Intracellular accumulation

SHORT NOTES:

1. Difference between necrosis and apoptosis.
2. Free radical cell injury
3. Radiation injury
4. Difference between dystrophic and metastatic calcification
5. Difference between hypertrophy and hyperplasia

SHORT ANSWERS:

1. Tigered effect / tabby cat heart
2. Role of sirtuins in cellular aging
3. Atrophy
4. Metaplasia

UPDATES

1. NECROSIS

DEFINITION

Necrosis is defined as the death of tissue due to intracellular proteins and enzymatic digestion of lethally injured cell, usually accompanied with inflammatory reaction

CAUSES

- Hypoxia
- Physical agents
- Chemical agents
- Microbial
- Immunological

MICROSCOPIC FINDINGS

- Increased eosinophilia
- Glassy, homogenous

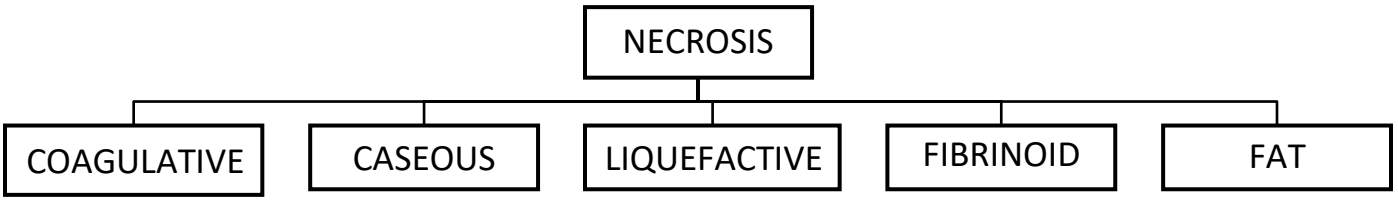
Dead cells replaced by myelin figures (phospholipids) → Phagocytosed by other cells → Fatty residues calcified

- Cells characterized by discontinuity in plasma and organelle membranes
- Dilated mitochondria

NUCLEAR CHANGES

- **KARYOLYSIS** – DNA loss due to DNAse and RNAse
- **PYKNOSIS** – nucleus condenses, clumping of chromatin
- **KARYORRHEXIS** – irreversible fragmentation of the nucleus

TYPES OF NECROSIS *(any type can be asked separately as 4m)*



COAGULATIVE NECROSIS

- Architecture of dead tissue is preserved
- Firm texture
- **Denaturation** of structural proteins and enzymatic proteins
- Dead cells removed by phagocytosis
- **Ischemia** due to obstruction leads to this necrosis (except brain)
- Area known as **infarct,** common in – heart, kidney, spleen
- **Microscopically** nucleus and cytoplasmic details lost **(tombstones),** swollen cell.

CASEOUS NECROSIS

- Occurs in **tuberculous** infections
- Cheese like appearance (friable, white)
- Combines features of both **coagulative and liquefactive necrosis**
- **Microscopically,** structureless lysed cells collection, amorphous debris enclosed within inflammatory border – **Granuloma**
- Common sites – lung, lymph nodes, intestines, TB of any organ.

LIQUEFACTIVE NECROSIS

- Dead cells **digested**
- Seen as **Ischemic injury in CNS**, bac/fungal infections
- Tissue degradation by hydrolytic enzymes – **liquid viscous mass**
- Creamy yellow – dead leukocytes –**PUS**
- **Microscopically,**
  - → Cyst contains debris and macrophages.
  - → Gliosis, inflammatory cells and fibroblasts.

FIBRINOID NECROSIS

- Seen in **immune reactions** involving blood vessels
- **Ag – Ab complex** lodged in wall of arteries
- This with **fibrin**, leaks out, forms bright pink amorphous stain in H&E staining.
- **Microscopically,** hyaline like deposit on wall, nuclear debris of neutrophils

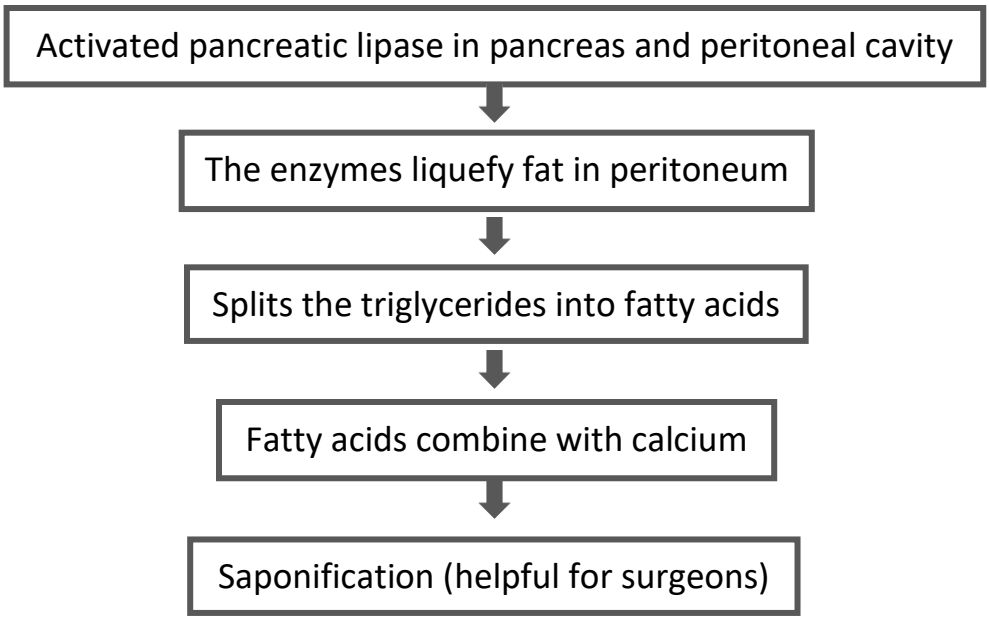
FAT NECROSIS

- Basically, focal areas of **fat destruction**
- Seen in **acute pancreatitis**

  Microscopically,

  → Necropsied fat cells are cloudy, enclosed by inflammatory cells

  → **Ca soaps** – amorphous, granular, basophilic



TRAUMATIC FAT NECROSIS:

- Necrosis that occurs in fatty tissue (e.g., female breast tissue, abdomen) as a result of blunt trauma or surgery
- Unlike pancreatic fat necrosis, it is **not** enzyme-mediated.

2. APOPTOSIS

DEFINITION

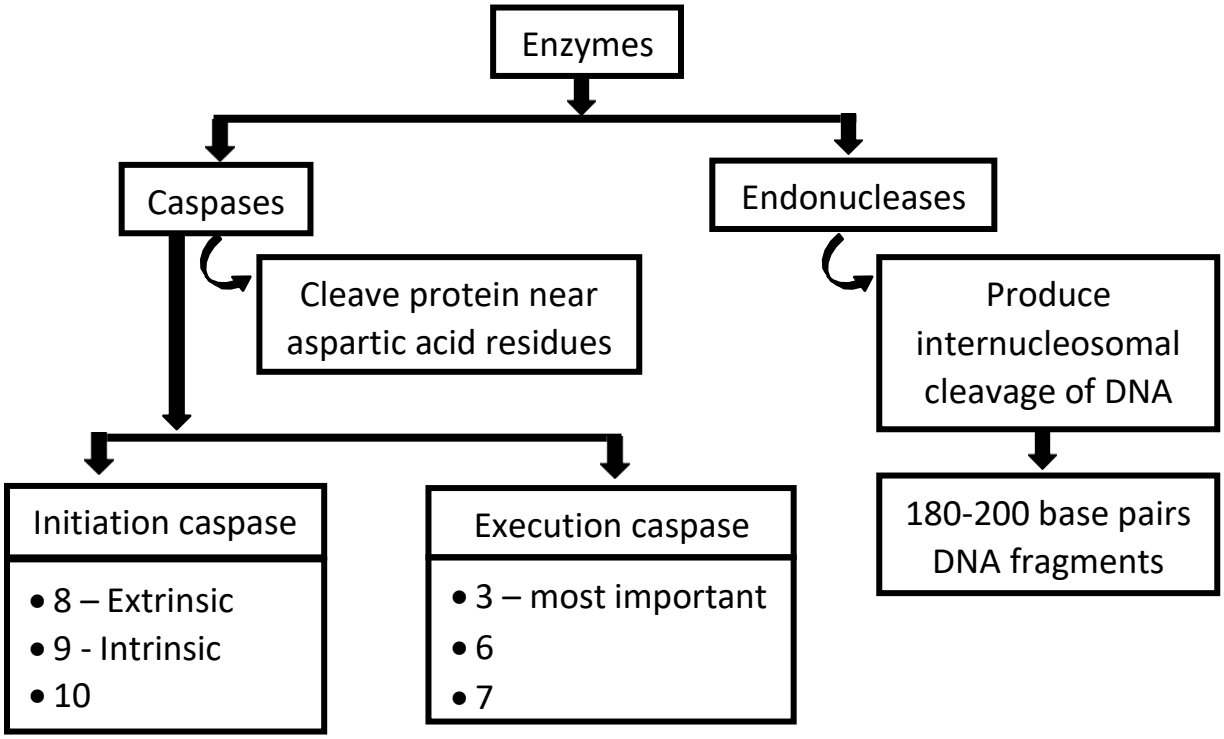
Apoptosis is a pathway of cell death that is induced by a **tightly regulated suicide program** in which cells destined to die activate intrinsic enzymes that degrade the cells' own nuclear DNA and nuclear and cytoplasmic proteins.

- It is a **genetically programmed cell death**
- It is an active process

CAUSES

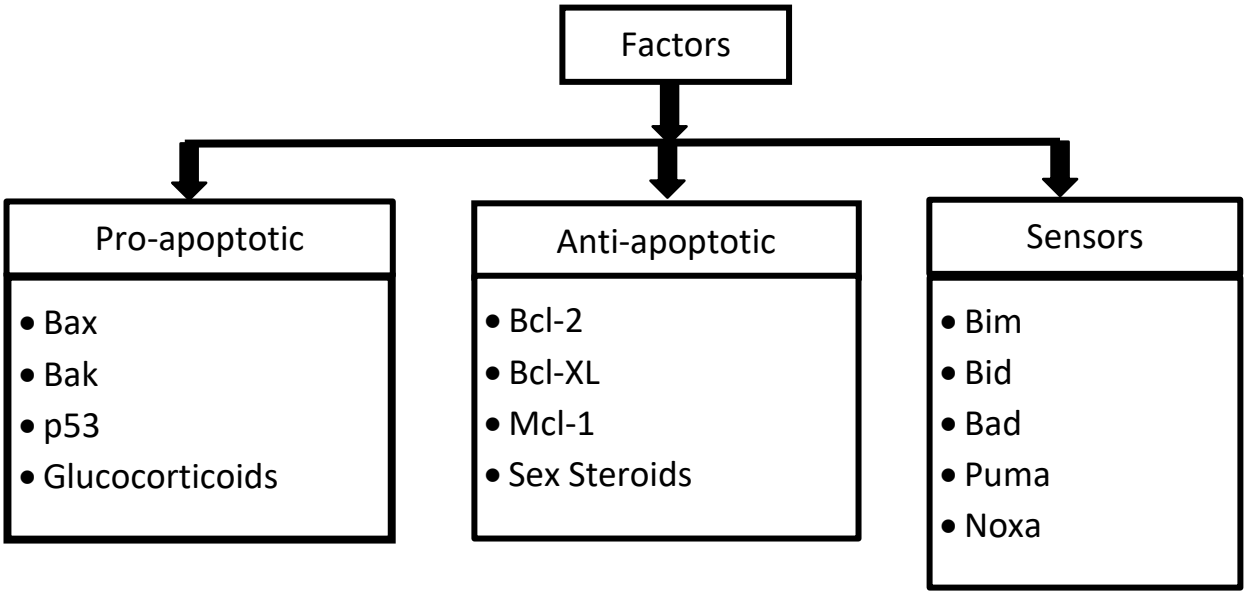
PHYSIOLOGICAL	PATHOLOGICAL
To eliminate cells that are no longer needed by the body and to maintain a steady number of various cell populations in tissues **Examples:** **Embryogenesis**: Destruction of cells during Implantation, Organogenesis, etc. **Involution of hormone dependent tissue upon hormone withdrawal** Endometrial cell breakdown during the menstrual cycle, **Cell loss in proliferating cell populations** Immature lymphocytes in the bone marrow **Elimination of potentially harmful self-reactive lymphocytes** Before or after they have completed their maturation **Death of host cells after their function**: Neutrophils in an acute inflammatory response	To remove cells that are injured beyond repair, so as to prevent eliciting host reaction and limit the collateral damage **Examples:** **DNA damage:** Either directly or by production of free radicals (radiation, cytotoxic drugs, hypoxia) **Accumulation of misfolded proteins:** Due to mutation in proteins / extrinsic factor (free radical injury) Accumulation in ER → **ER stress** → Apoptosis **Viral infection:** Loss of infected cells due to apoptosis induced by the virus (as in adenovirus and HIV infections) **Duct obstruction – atrophy of glands:** pancreas, parotid and kidney **Retinitis pigmentosum:** apoptosis of rod pigment **Transplant rejection**: Graft vs Host reaction

ENZYMES INVOLVED IN APOPTOSIS



- Caspases exist as pro-caspases (zymogen) → enzymatic cleavage → active caspase
- Presence of cleaved active caspase → marker of cells undergoing apoptosis

FACTORS INVOLVED IN APOPTOSIS *(can be asked in 1m)*

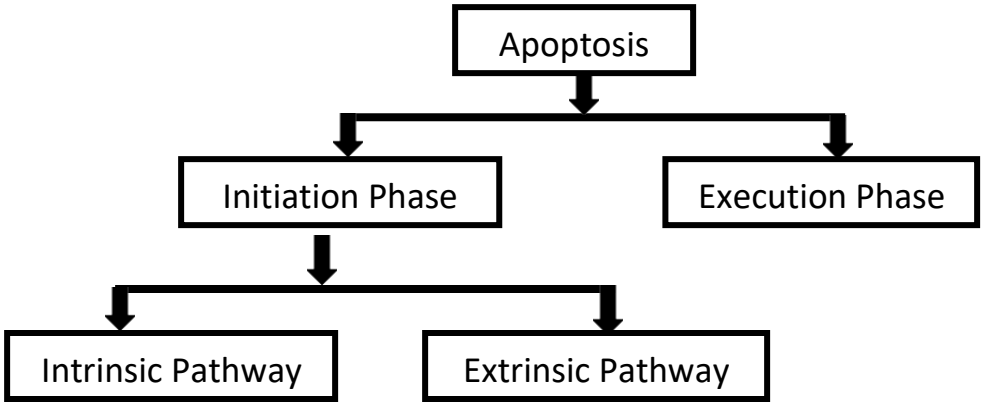


- **Pro-apoptotic:**
  - They have four BH (Bcl-2 Homology) domains – **BH1-4**
  - Upon activation, they oligomerize with outer mitochondrial protein and promote mitochondrial outer membrane permeability.

- **Anti-apoptotic:**
  - They have all four BH domains
  - They reside in
    - Outer mitochondrial membrane
    - Plasma membrane
    - ER membrane
  - They prevent leakage of death-inducing proteins (cytochrome C) by keeping mitochondrial membrane intact

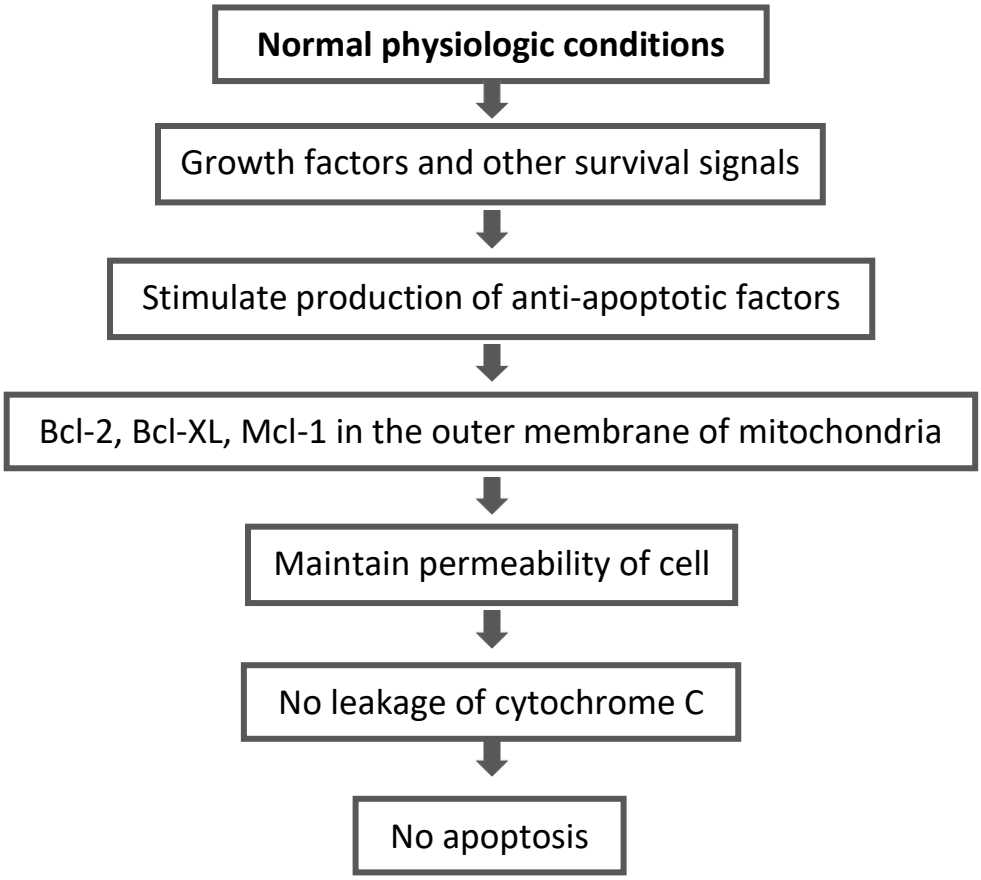
- **Sensors:**
  - They contain a single BH domain (third domain)
  - They are also called BH3-only proteins
  - They act as sensors of cellular stress and damage
  - They regulate the balance between pro- and anti-apoptotic factors
  - Thus, they are called arbiters of apoptosis

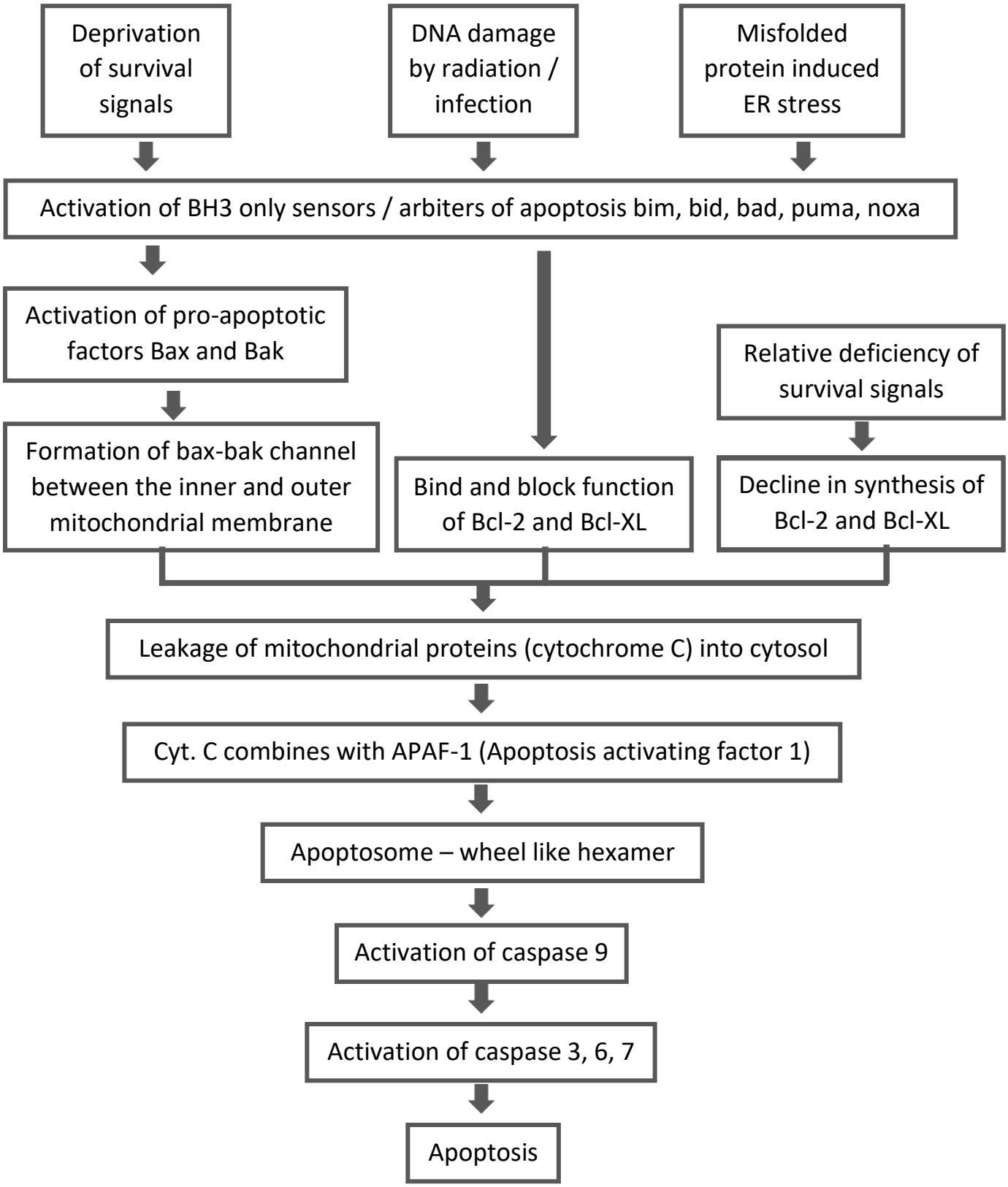
MECHANISM OF APOPTOSIS *(can be asked separately as 4m)*



INTRINSIC PATHWAY

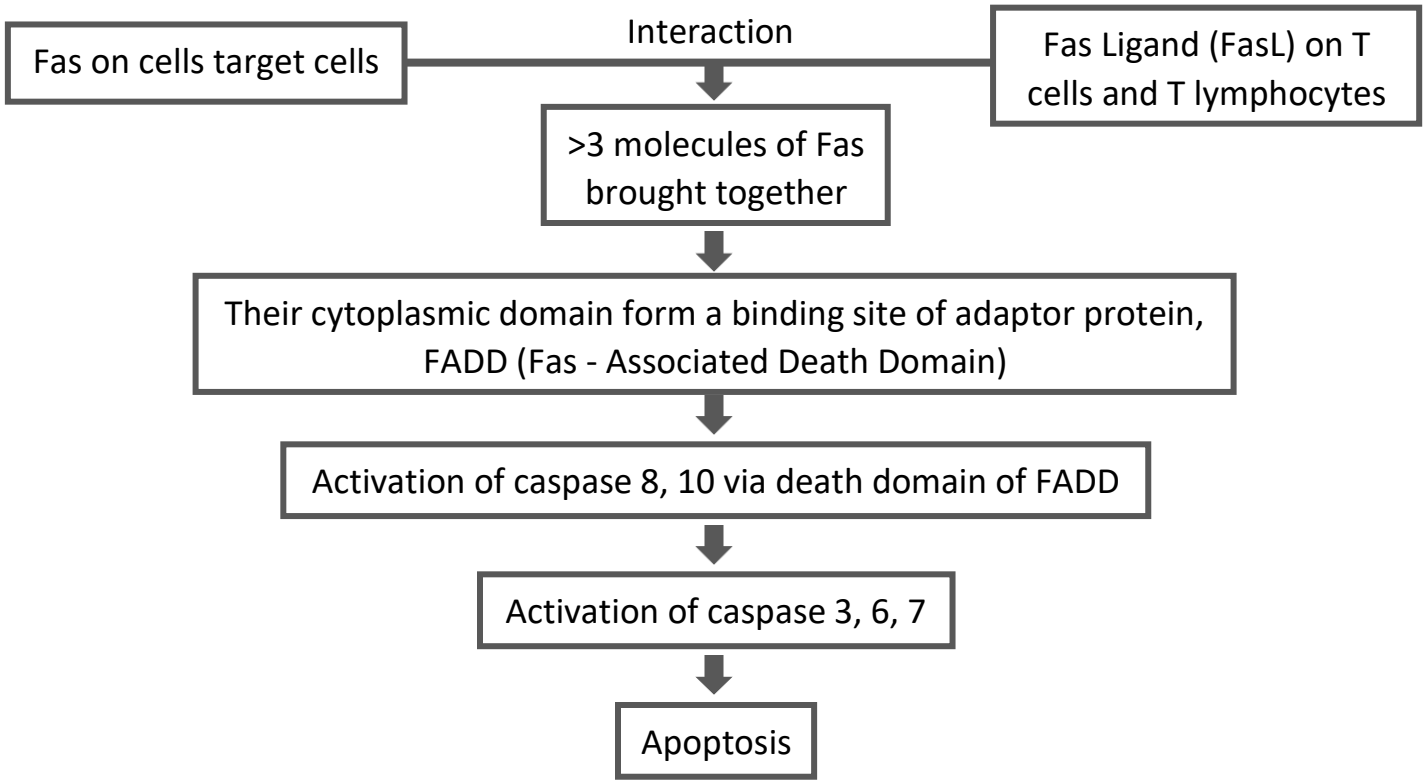
- It is also known as **Mitochondrial Pathway**
- It is the major mechanism of apoptosis in mammalian cells
- The most common organ affected in apoptosis is mitochondria



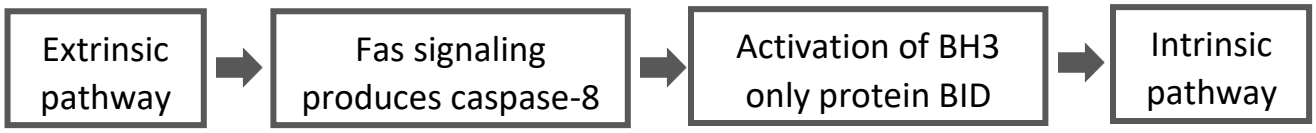

Deprivation of survival signals
DNA damage by radiation / infection
Misfolded protein induced ER stress
Activation of BH3 only sensors / arbiters of apoptosis bim, bid, bad, puma, noxa
Activation of pro-apoptotic factors Bax and Bak
Relative deficiency of survival signals
Formation of bax-bak channel between the inner and outer mitochondrial membrane
Bind and block function of Bcl-2 and Bcl-XL
Decline in synthesis of Bcl-2 and Bcl-XL
Leakage of mitochondrial proteins (cytochrome C) into cytosol
Cyt. C combines with APAF-1 (Apoptosis activating factor 1)
Apoptosome – wheel like hexamer
Activation of caspase 9
Activation of caspase 3, 6, 7
Apoptosis

EXTRINSIC PATHWAY

- It is also known as **Death receptor mediated pathway**
- Initiated by engagement of plasma membrane death receptors
- **Death receptors:**
  - Member of TNF receptor family
  - Cytoplasmic domain / Death domain:
    - Involved in protein-protein interactions
    - Essential for delivering apoptotic signals
    - E.g.: type 1 TNF receptor (TNFR1); Fas (CD95)



Intrinsic + Extrinsic pathway:



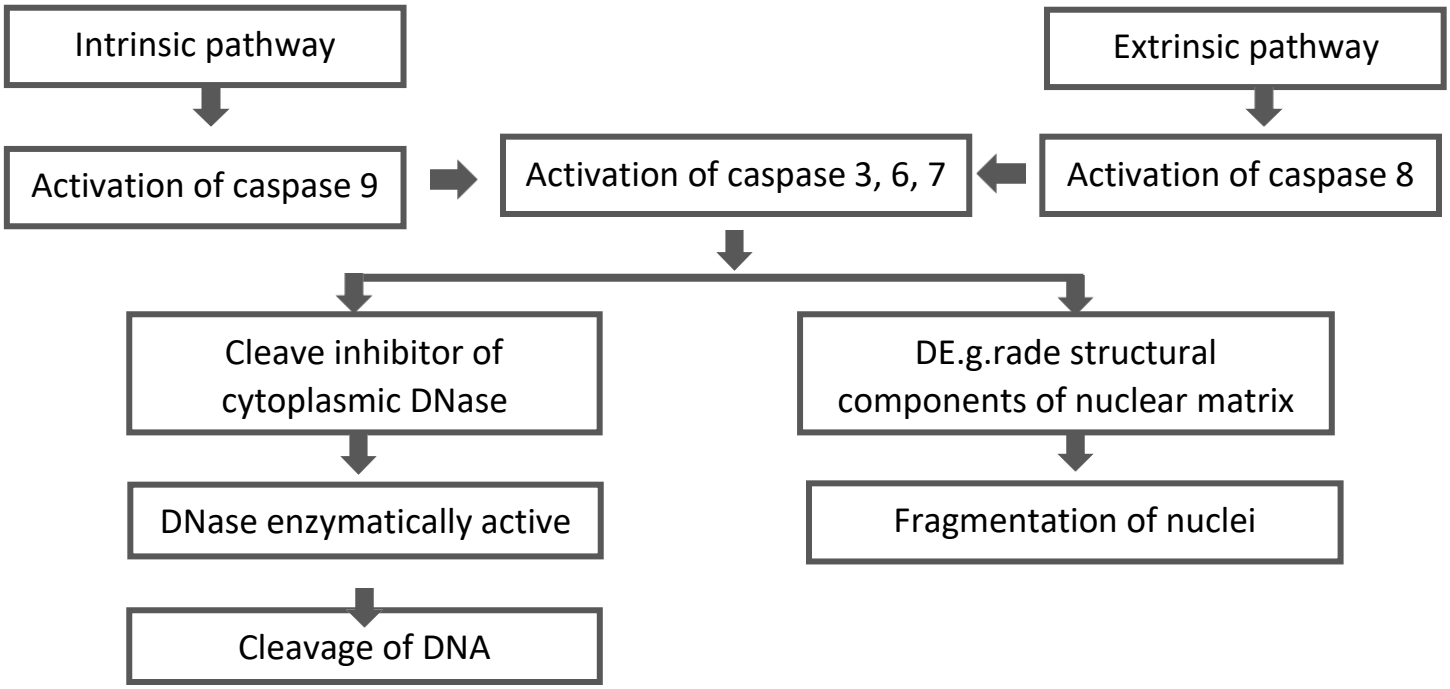
- It occurs in hepatocytes and pancreatic β cells
- The combined activation of both pathways delivers a fatal blow to the cells

INHIBITORS

- **Inhibitor of extrinsic pathway** – FLIP
  - inactivates procaspase 8
  - lacks protease domain→can't cleave and activate caspase

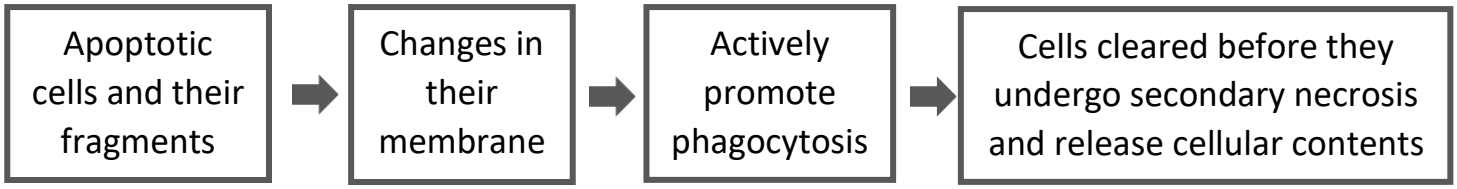
- **Inhibitor of intrinsic pathway** – IAP (physiological Inhibitors of Apoptosis)
  - Inactivates procaspase-9 and executioner caspases like caspase-3
  - Neutralized by Smac/Diablo which enters the cytoplasm through Bax-Bak channel

EXECUTION PHASE

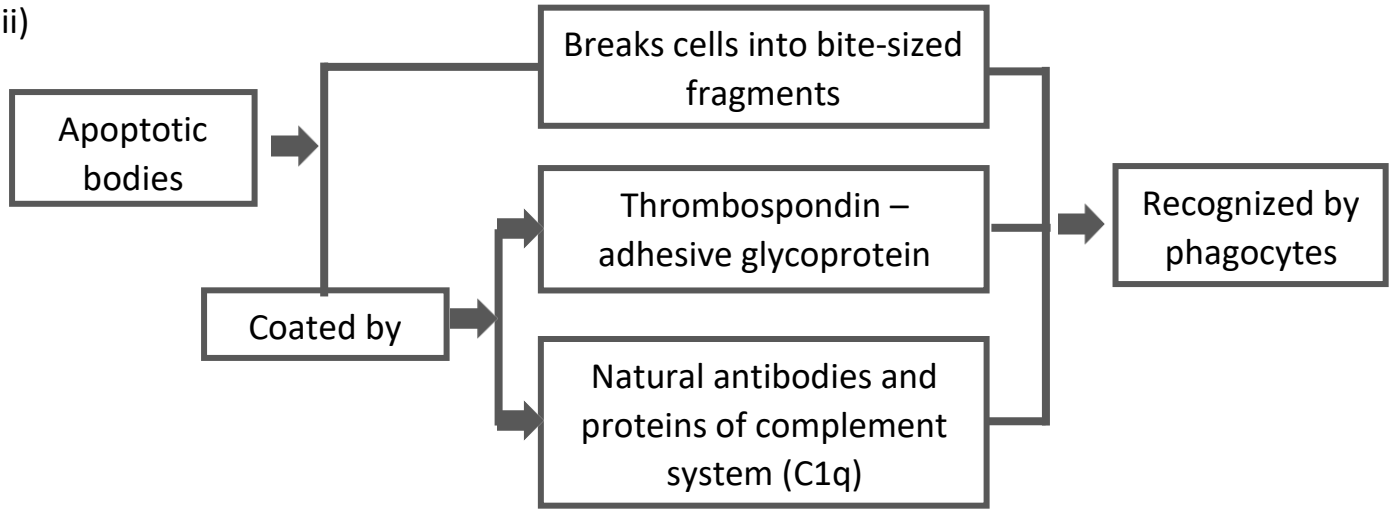


REMOVAL OF DEAD CELLS

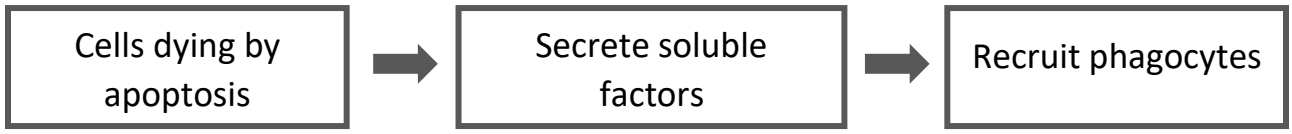
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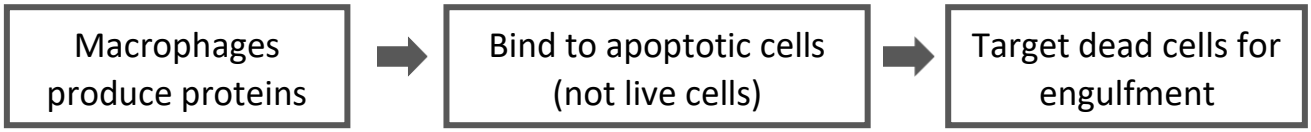
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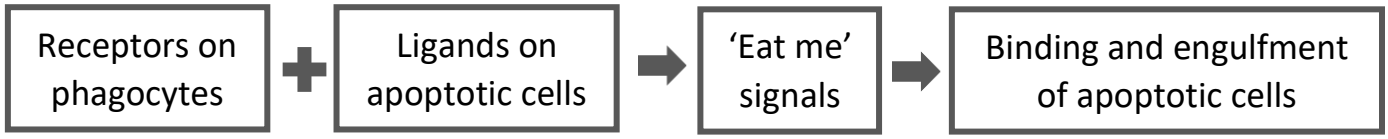
iii)



iv)



v)

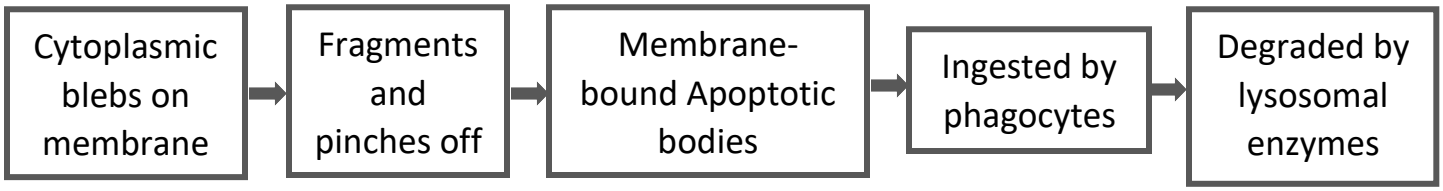


MORPHOLOGICAL FEATURES OF APOPTOSIS

- **Cell shrinkage:**
  - Earliest feature.
  - The cytoplasm is dense.
  - Organelles are relatively normal and more tightly packed.

- **Chromatin condensation:**
  - Most characteristic feature
  - The chromatin aggregates peripherally under the nuclear membrane,
  - The aggregates form dense masses of various shapes and sizes.
  - The nucleus may break up, producing two or more fragments

- **Formation of cytoplasmic blebs and apoptotic bodies:**



  - **Apoptotic bodies:** composed of cytoplasm and tightly packed organelles with or without nuclear fragments
  - Plasma membrane remains intact → no leakage of enzymes
  - At later stages, plasma membrane becomes permeable to normally retained solutes

- **Histologic features:**
  - Apoptotic cell appears as a round or oval mass of intensely eosinophilic cytoplasm with fragments of dense nuclear chromatin (Fig. 2-22A).
  - Apoptosis does not elicit inflammation → difficult to detect histologically

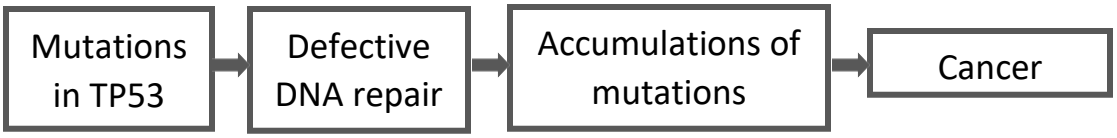
BIOCHEMICAL FEATURES

- ATP is required (energy dependent process)
- There is a transient loss of mitochondrial membrane potential
- pH of the cell is acidic
- It is caspase dependent
- There is exteriorization of Phosphatidyl Serine (PD) from inner to outer leaflet of plasma membrane – Phosphatidyl Serine Flip

- There is non-random mono and oligonucleosomal length fragment of DNA ➡ step ladder pattern in agarose gel electrophoresis
- Here, the DNA fragmentation is pre-lytic

DYSREGULATED APOPTOSIS *(can be asked separately as 4m)*

➢ **Defective apoptosis and increased cell survival.**

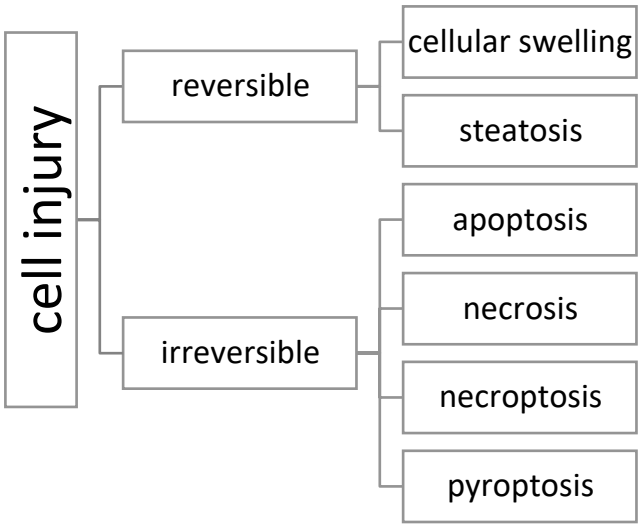


- Autoimmune diseases: Failure to eliminate
  - Lymphocytes ➡react against self-antigens
  - Dead cells ➡ source of self-antigens.

➢ **Increased apoptosis and excessive cell death**:

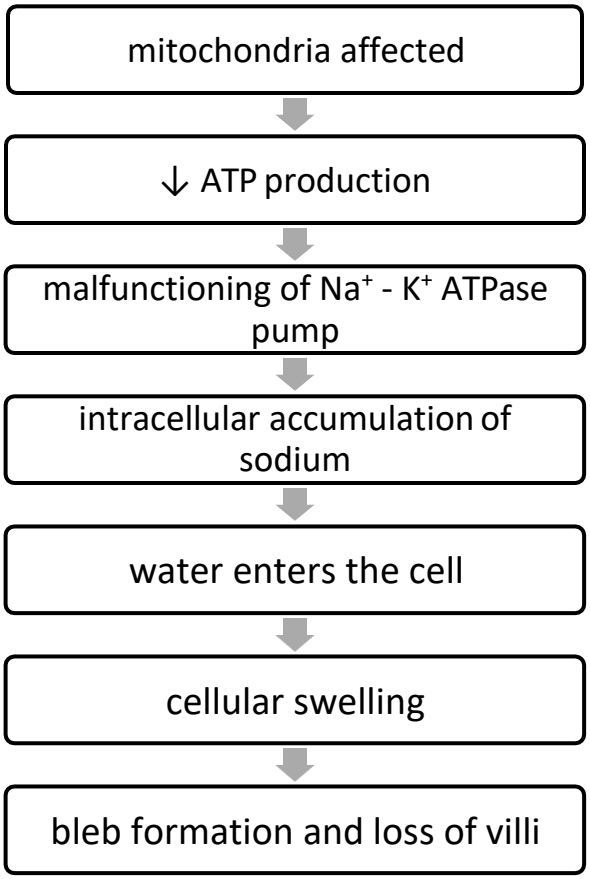
- Neurodegenerative diseases:
  - Loss of specific set of neurons
  - Mutations and misfolded proteins → Apoptosis
- Ischemic injury:
  - Myocardial infarction
  - Stroke
- Viral infections: Death of virus-infected cells.

3. TYPES OF DEGENERATION (CELL INJURY)



REVERSIBLE CELL INJURY:

CELLULAR SWELLING:



- Entry of water into phospholipid layer leading to partial denaturation, leading to extracellular curling of membrane upon itself called **myelin figures.**
- ER-decreased protein synthesis and accumulation of misfolded proteins
- **Lactic acidosis** due to decreased glycolysis

FATTY CHANGE: (STEATOSIS) *(explained in detail in Q4)*

→ Abnormal accumulation of **triglycerides within parenchymal cells**

→ The most common organ affected in **liver** as it is more involved in fat metabolism

→ Heart, muscle and kidney are also affected

→ Causes of steatosis are certain toxins, protein malnutrition, diabetes mellitus, obesity, and anoxia.

FAT STAINS

→ Sudan Black B stains black
→ Oil red O stains reddish-orange
→ Sudan 3 stains red to yellowish

IRREVERSIBLE INJURY:

APOPTOSIS: CASP mediated programmed cell death

NECROSIS:

- death of tissue irreversibly
- injury causing agent present for long time
- **types**-coagulative, liquefactive, caseous, fat necrosis, fibrinoid necrosis and gangrenous necrosis

NECROPTOSIS: CASP independent programmed cell death, also called as programmed necrosis

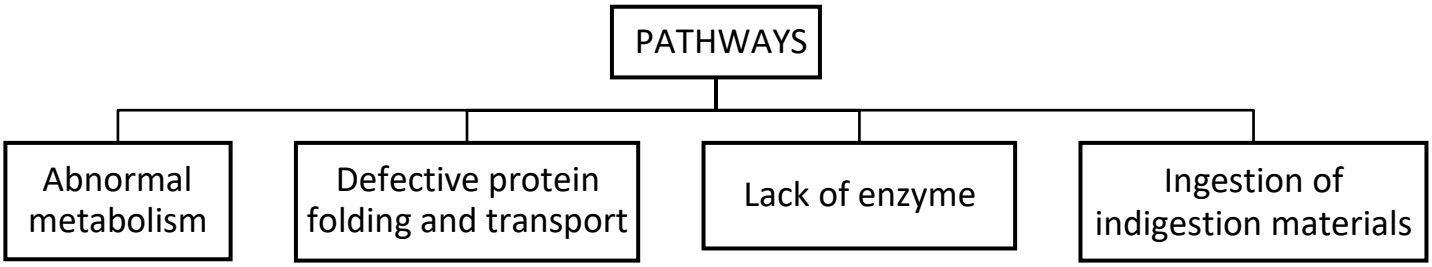
PYROPTOSIS: highly inflammatory form of programmed cell death

4. INTRACELLULAR ACCUMULATIONS

- Intracellular accumulations are one of the manifestations of metabolic derangement of cells
- They may be harmless or associated with varying degree of injury
- The substances may be accumulated in **cytoplasm, organelles (especially lysosomes), or nucleus**
- They may be either synthesized by affected cells or produced elsewhere
- If the overload is controlled / stopped → the accumulation is reversible.

Inherited storage disorders → Accumulation is progressive → Cellular injury due to overload → Death of tissue and patient

PATHWAYS OF INTRACELLULAR ACCUMULATION



Abnormal metabolism:

- Defects in mechanism of packaging and transport → Inadequate removal of normal substance
- E.g.: Fatty change (steatosis) in liver

Defect in protein folding, transport:

- Protein folding / Packaging / Transport / Secretion → Genetic / Acquired defects → Accumulation of abnormal endogenous substrate
- E.g.: Certain mutated forms of α1-antitrypsin deficiency

Lack of enzyme:

- Inherited enzyme deficiencies → Failure of degradation of metabolites → Accumulation of endogenous materials
- E.g. Lysosomal Storage Disorders

Ingestion of indigestible materials:

- Enzymatic machinery to degrade substrate / Ability to transport to other sites → Absence → Deposition and accumulation of abnormal exogenous substrate
- E.g.: Accumulation of Carbon and silica particles

LIPIDS *(can be asked separately as 4m)*

- All major classes of lipids accumulate in cells:
  - **Triglycerides**
  - **Cholesterol / cholesterol esters**
  - **Phospholipids.**
- Phospholipids are components of the myelin figures found in necrotic cells.
- Abnormal complexes of lipids and carbohydrates accumulate in the lysosomal storage diseases.

➢ Steatosis (Fatty Change)

- The terms steatosis and fatty change describe abnormal accumulations of triglycerides within parenchymal cells**.**
- Fatty change is often seen in the liver because it is the major organ involved in fat metabolism (Fig. 2-30),
- Fatty change also occurs in heart, muscle, and kidney.
- Causes of steatosis:
  - Toxins
  - Protein malnutrition
  - Diabetes mellitus
  - Obesity
  - Anoxia

- In developed nations, the most common causes of fatty liver:
  - Alcohol abuse
  - Nonalcoholic fatty liver disease - associated with diabetes and obesity.

➢ Cholesterol and Cholesterol Esters

- The cellular metabolism of cholesterol is tightly regulated such that the cells use cholesterol for cell membranes synthesis.
- Accumulation of cholesterol or cholesterol esters are manifested histologically by intracellular vacuoles.
- **Atherosclerosis:**
  - Here, the smooth muscle cells and macrophages within the intima of the aorta and large arteries are filled with lipid vacuoles – Foam cells (has a foamy appearance
  - Aggregates of foam cells produce the yellow cholesterol-laden atheromas - characteristic of the disease.
  - The foam cells may rupture, releasing lipids into the extracellular space.
  - The extracellular cholesterol esters crystallize into long needles, producing distinctive clefts in tissue sections.

- **Xanthomas:**
  - Intracellular accumulation of cholesterol within macrophages → characteristic of acquired and hereditary hyperlipidemic states**.**
  - Clusters of foamy cells are found in the subepithelial connective tissue of the skin and in tendons, producing tumorous masses known as xanthomas.

- **Cholesterolosis:**
  - Focal accumulations of cholesterol-laden macrophages in the lamina propria of the gallbladder

- **Niemann-Pick disease, type C:**
  - Mutations affecting an enzyme involved in cholesterol trafficking → cholesterol accumulation in multiple organs

PROTEINS

- They appear as rounded, eosinophilic droplets, vacuoles, or aggregates in the cytoplasm.
- They may be amorphous, fibrillar, or crystalline based on electron microscopy.
- Protein can also be deposited in extracellular spaces (as in amyloidosis)

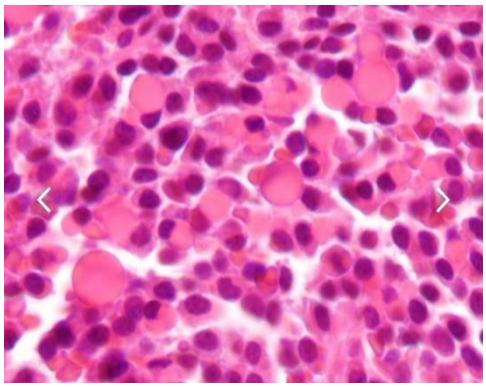
Excesses of proteins within the cells sufficient to cause morphologically visible accumulation have diverse causes.

➢ **Reabsorption droplets in proximal renal tubules**:

- These droplets are seen in renal diseases associated with proteinuria.
- Physiological: Small amount of protein filtered through glomerulus → reabsorbed by pinocytosis in proximal tubule.
- Disorders with heavy protein leakage → Increased reabsorption of the protein into vesicles.
- These proteins appear as pink hyaline droplets within the cytoplasm of the tubular cell
- The process is reversible; if the proteinuria diminishes, the protein droplets are metabolized and disappear.

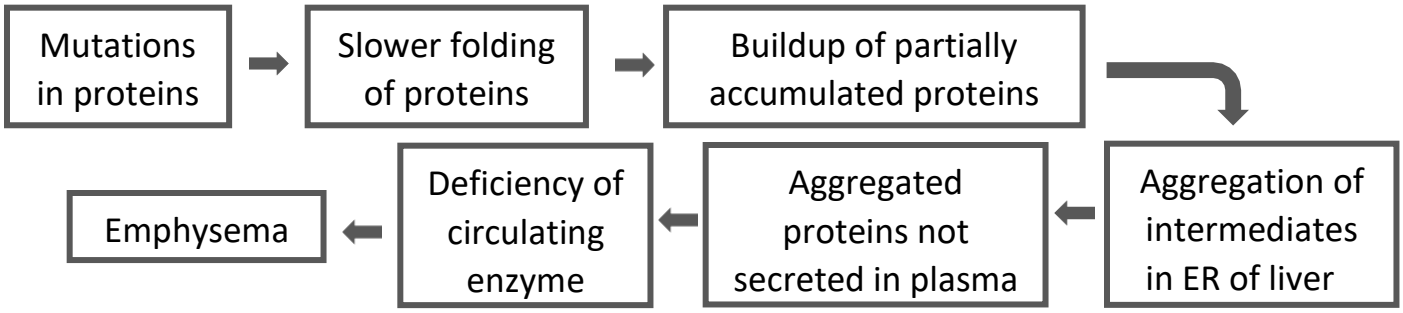
➢ **Russell bodies**: *(can be asked separately as 1m)*

- They are large, homogenous, eosinophilic inclusions within hugely distended Endoplasmic Reticulum
- They are seen in certain plasma cells involved in active synthesis of immunoglobulins.
- Here the accumulated protein (immunoglobulin) is a normally secreted protein that are produced in excessive amounts



➢ **Defective intracellular transport and secretion of critical proteins:**

- α1-antitrypsin deficiency:



- Here the pathology results from:
  - Loss of protein function
  - ER stress caused by misfolded proteins → Apoptosis

➢ **Accumulation of cytoskeletal proteins.**

- Types of cytoskeletal proteins:
  - Microtubules
  - Thin actin filaments
  - Intermediate filaments
  - Thick myosin filaments
- Intermediate filaments provide a flexible intracellular scaffold which organizes the cytoplasm and resists forces applied to the cell.
- The Intermediate filaments are divided into five classes:

Keratin filaments	Characteristic of epithelial cells
Neurofilaments	Neurons
Desmin filaments	Muscle cells
Vimentin filaments	Connective tissue cells
Glial filaments	Astrocytes

- Accumulations of keratin filaments and neurofilaments are associated with certain types of cell injury.
- Alcoholic hyaline:
  - Eosinophilic cytoplasmic inclusion in liver cells
  - Characteristic of alcoholic liver disease
  - Composed predominantly of keratin
- Neurofibrillary tangle:
  - Contains neurofilaments and other proteins
  - Found in Alzheimer's disease

➢ **Aggregation of abnormal proteins:**

- They are also known as proteinopathies or protein-aggregation diseases.
- Here, abnormal or misfolded proteins deposit in tissues and interfere with normal functions.
- It can deposit intracellularly / extracellularly / both
- The deposited proteins induce pathologic changes either directly or indirectly
- E.g.: Certain forms of amyloidosis

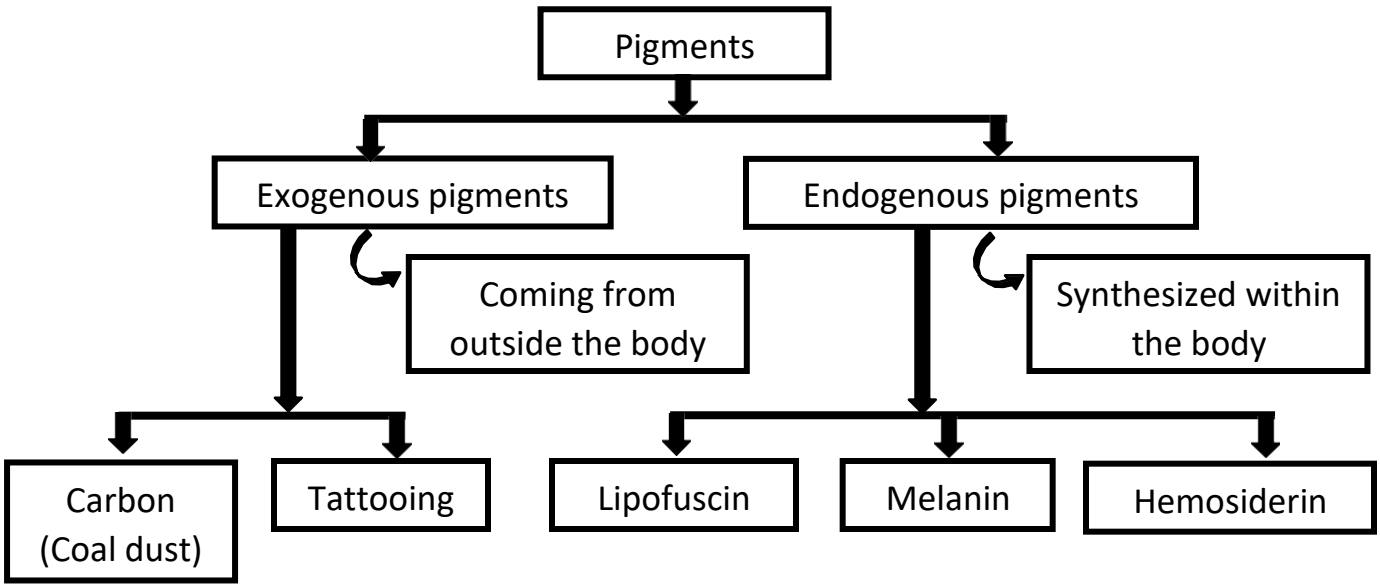
GLYCOGEN

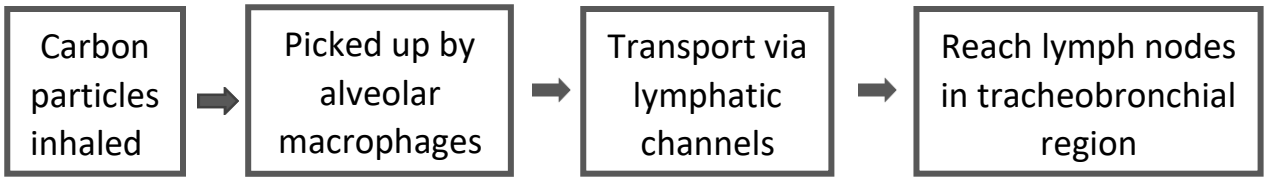
- It is seen in patients with abnormality with glucose or glycogen metabolism
- Glycogen masses appear as clear vacuoles within the cytoplasm
- It is readily identified when tissues are fixed in absolute alcohol (since glycogen dissolves in aqueous fixatives)
- Staining with Best carmine / PAS reaction → Imparts rose-to-violet color to the glycogen
    - Diastase digestion of a parallel section before staining serves as a control by hydrolyzing the glycogen.

- **Diabetes Mellitus**: Here, glycogen is found in
    - Renal tubules – **Armanni Ebstein lesion**
    - Liver parenchyma
    - β cells of the islets of Langerhans
    - Cardiac myocyte

PIGMENTS *(can be asked separately as 4m)*

Pigments are colored substances, some of which are normal constituents of the cell (melanin) whereas others are abnormal and accumulate under special circumstances.



- **Carbon (coal dust)**
  - It is the most common exogenous pigment.
  - It is a ubiquitous air pollutant in urban areas.
  - 

  - **Anthracosis:** Blackening of lung tissue and lymph nodes due to accumulation of carbon.
  - **Coal worker's pneumoconiosis:** Emphysema due to the fibroblastic reaction induced by aggregates of carbon dust in coal miners

- **Tattooing**
  - It is a form of localized, exogenous pigmentation of the skin
  - The inoculated pigments are phagocytosed by dermal macrophages where they remain forever.
  - These pigments do not usually evoke any inflammatory response

- **Lipofuscin/ lipochrome/ wear & tear pigment** *(important)*
  - Insoluble endogenous pigment
  - Contains polymers of lipids and phospholipids complexed with proteins
  - Produced by lipid peroxidation of membranes
  - Not injurious to cell or cellular functions.
  - Telltale sign of free radical injury and lipid peroxidation**.**
  - Seen in cells undergoing slow regressive changes, specifically in the liver and heart (brown atrophy of liver and heart)
  - It is also seen in patients in severe malnutrition and cancer cachexia.
  - Yellow brown, finely granular perinuclear pigment

- **Melanin**
  - It is an endogenous, brown-black pigment
  - It is formed by the enzyme tyrosinase which catalyzes the oxidation of tyrosine to dihydroxyphenylalanine in melanocytes
  - Marker: HMB-45, Melan-A, S-100
  - **Special Stain:** Masson Fontana

- **Hemosiderin**
  - It is a hemoglobin-derived, golden yellow-to-brown, granular or crystalline pigment
  - It is deposited in conditions with iron overload
  - In cells, iron is stored in association with apoferritin to form ferritin micelles.
  - When there is a local or systemic excess of iron, ferritin forms hemosiderin granules, which are easily seen with the light microscope.
  - **Normal condition:**
    - Small amounts of hemosiderin can be seen in the mononuclear phagocytes of the bone marrow, spleen, and liver.
  - **Local excess**: common bruise
  - **Systemic excess / Hemosiderosis:** Due to
    - Hemochromatosis - increased absorption of dietary iron due to an inborn error of metabolism
    - Hemolytic anemias - premature lysis of red cells leads to release of abnormal quantities of iron
    - Repeated blood transfusions - transfused red cells constitute an exogenous iron load
  - **Special Stain:** Prussian blue / Perl's stain

SHORT NOTES

1. DIFFERENCE BETWEEN NECROSIS AND APOPTOSIS

FEATURE	NECROSIS	APOPTOSIS
DEFINITION	Cell death along with degeneration of tissues by hydrolytic enzymes	Programmed and coordinated cell death.
CAUSATIVE AGENTS	Hypoxia, Toxins	Physiological and pathological process
MARKERS	No marker	Annexin V, CD 95

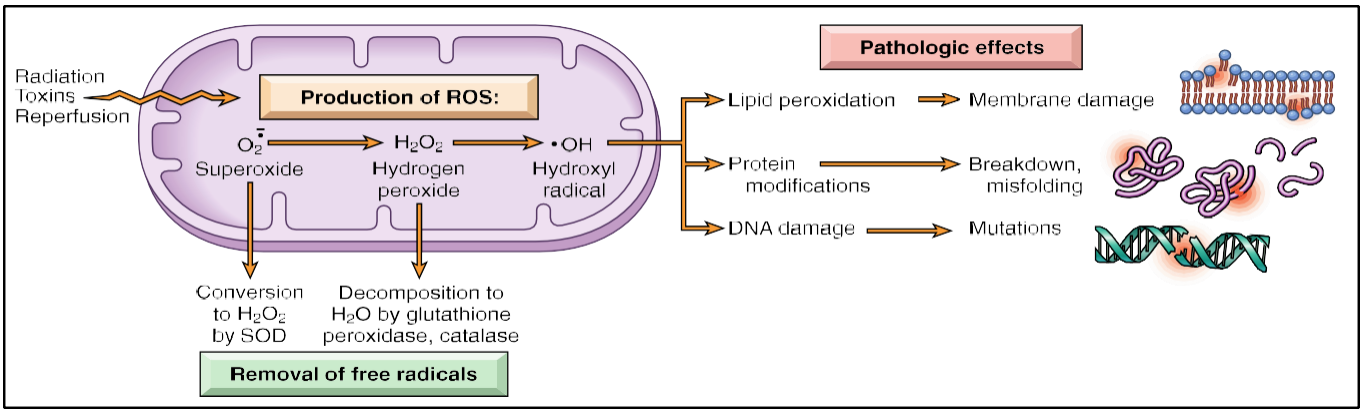
MORPHOLOGY	Inflammatory reaction always present	No inflammatory reaction
	Death of many adjacent cells	Death of single cells
	Cell swelling initially	Cell shrinkage
	Membrane disruption	Cytoplasmic blebs on membrane
	Damaged organelles	Apoptotic bodies
	Nuclear disruption	Chromatin condensation
	Phagocytosis of cell debris by macrophages	Phagocytosis of apoptotic bodies by macrophages
MOLECULAR CHANGES	Lysosomal breakdown with liberation of hydrolytic enzymes	Lysosome and other organelles intact
	Initiated by: • Ischemic hypoxia • Chemicals • Physical agents, etc.	Initiation by: • Loss of signals of normal cell survival • Action of agents injurious to cell
	Cell death by: • ATP depletion • Membrane damage • Free Radical injury	Triggered by: • Intrinsic/Mitochondrial pathway (pro- and anti-apoptotic members of Bcl-2 family) • Extrinsic/Death receptor mediated pathway (TNF-R1, Fas, Fas-L) • Finally activated by caspases
BIOCHEMICAL ASPECTS	Energy independent process (ATP not required)	Energy dependent process (ATP required)
	Permanent loss of mitochondrial membrane potential	Transient loss of mitochondrial membrane potential
	Caspases independent	Caspase dependent
	pH of the cell is unchanged	pH of the cell is acidic
	Phosphatidyl Serine remain unchanged	Exteriorization of Phosphatidyl Serine (PD) from inner to outer leaflet of plasma membrane
	Random digestion of DNA	Non-random mono and oligonucleosomal length fragment of DNA
	Smear pattern in agarose gel electrophoresis	Step ladder pattern in agarose gel electrophoresis
	Post-lytic DNA fragmentation	Pre-lytic DNA fragmentation

2. FREE RADICAL CELL INJURY

- Cell injury by free radicals, especially **reactive oxygen species (ROS)** – present in
  - → **Radiation**
  - → **Ischemia reperfusion injury**
  - → **Ageing**
  - → **Microbial killing by phagocytes**.

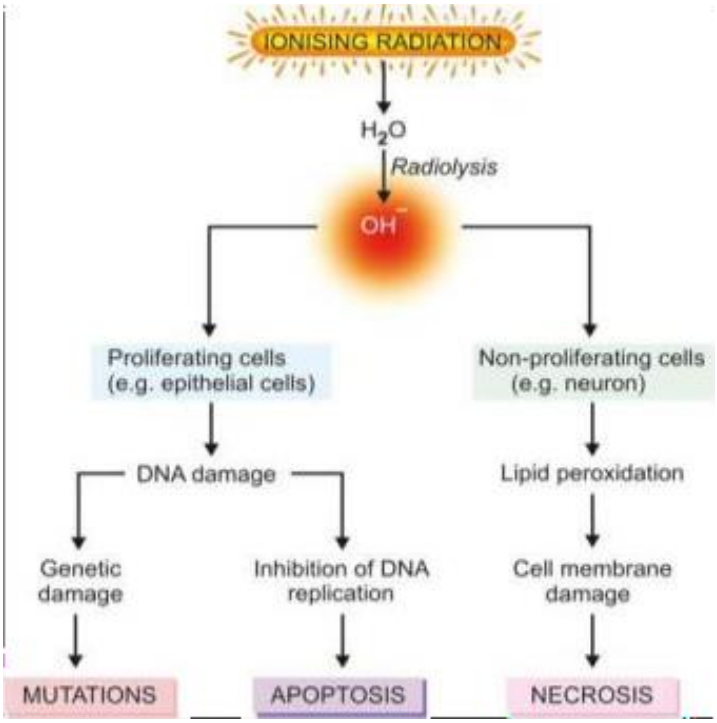
- Free radicals– species with **single unpaired electron in the outer orbit**
- **Energy from this unstable configuration is used in reacting with organic & inorganic molecules**
- Initiate **autolytic reactions**, where the reactants they react with, turn into free radicals themselves causing a chain of damage.
- ROS produced normally during mitochondrial respiration and energy degeneration, but usually destroyed by the cell

When increased ROS or defect in the destruction → Accumulation of free radicals → **OXIDATIVE STRESS**



3. RADIATION INJURY

- Result of direct formation of **hydroxyl radicals from radiolysis of water**
- Radiation injury to human by **accidental or therapeutic exposure** is of importance in **treatment of persons with malignant tumours** as well as may have carcinogenic influences



4. DIFFERENCE BTW DYSTROPHIC AND METASTATIC CALCIFICATION:

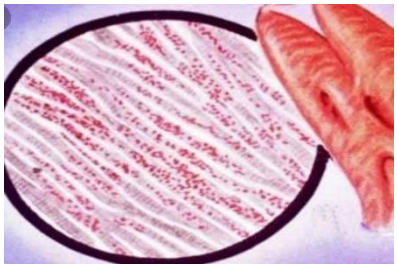
FEATURE	DYSTROPHIC CALCIFICATION	METASTATIC CALCIFICATION
DEFINITION	Deposits of calcium salts in dead and degenerated tissues	Deposition of calcium salts in normal tissues
CALCIUM METABOLISM	Normal	Deranged
SERUM CALCIUM LEVEL	Normal	Hypercalcemia
REVERSIBILITY	Mostly irreversible	Reversible on adequate metabolic disorder correction
CAUSES	Necrosis, infarcts, thrombi, atheromas, monkebergs sclerosis, etc.	Hyperparathyroidism, bony destructive lesions, hypervitaminosis D, etc.
PATHOGENESIS	↑ binding of phosphates with necrotic and degenerative tissue, which in turn binds to calcium forming calcium phosphates precipitates	↑ precipitates of calcium phosphate due to hypercalcemia at certain sites (lungs, stomach, etc.)

5. DIFFERENCE BETWEEN HYPERTROPHY AND HYPERPLASIA

FEATURE	HYPERTROPHY	HYPERPLASIA
DEFINITION	Increase in size of cells	Increase in number of cells
CELLS AFFECTED	Involves both cells – capable of dividing and non-dividing cells	Occurs in cells capable of synthesizing DNA and undergoing mitotic activity
MOLECULAR PATHOGENESIS	Increased synthesis of proteins by increased growth factors or stimulus which causes activation of signal transduction pathways	Increased proliferation of cells either due to increased growth factors or growth factor receptors which stimulates signal transduction pathways
MORPHOLOGY	Increase in cell size in nucleomegaly	Increases cell number with increased number of mitotic figures
ELECTRON MICROSCOPY	Increases number of organelles and increased synthesis of DNA, RNA and myofilaments	Increased DNA and typical mitosis
OUTCOME	Reversible on withdrawal of stimulus.	Regress after removal of stimulus. In case of neoplastic transformations, it will nor regress
EXAMPLES	**Physiological:** • Uterine myometrium in pregnancy • Breast during lactation • Skeletal muscles in body builders	**Physiological:** • Hormonal hyperplasia of breast during pregnancy and puberty • Compensatory hyperplasia of liver after partial hepatectomy
	Pathological: • Myocardial hypertrophy • Hypertrophy of smooth muscles at pylorus leading to stenosis	**Pathological:** • Endometrial hyperplasia and benign prostatic hyperplasia • Hyperplasia in skin warts

SHORT ANSWERS

1. TIGERED EFFECT / TABY CAT EFFECT



- It is usually caused due to **fatty degeneration of the myocardium.**
- **Yellow and white stripes alternately** resemble the **skin of tiger** and hence named as Tigered effect.

2. ROLE OF SIRTUINS IN CELL AGING

- The sirtuins (SIRTs) constitute a class of proteins with nicotinamide adenine dinucleotide-dependent deacetylase or **adenosine diphosphate-ribosyltransferase activity**.
- **Seven SIRT** family members have been identified in mammals
- SIRT1 and SIRT2 are localized in the nucleus and cytoplasm.
- SIRT3, SIRT4, and SIRT5 are mitochondrial
- SIRT6 and SIRT7 are nuclear
- SIRT proteins regulate diverse **cell functions and responses to stressors**.

3. ATROPHY

- It is defined as a reduction in the size of an organ or tissue due to decrease in cell size and number
- Types
  - → Physiological – atrophy of notochord, thyroglossal cyst during fetal development
  - → Pathological – immobilization, denervation atrophy, ↓ blood supply, etc.

4. METAPLASIA

→ It is a reversible change in which one differentiated cell type is replaced by another cell type.

→ e.g. normal ciliated columnar epithelium of trachea changes to stratified squamous epithelium in chronic smokers

(note: the differentiated cells don't undergo metaplasia, rather it's the undifferentiated cells that undergo metaplasia)

UPDATES FROM ROBBINS: 10TH EDITION:

1. Satellite DNA:

- It is a class of repetitive sequences in centromere that are organized in long arrays of tandemly repeated units (from 5bp to 5kb)
- It helps in spindle cell apparatus attachment
- It is also important in maintaining the dense, thickly packed organization of heterochromatin

2. Efferocytosis:

- It is the process of phagocytosis through the 'eat me signals' in apoptosis
- It is carried out by conventional phagocytes like macrophages and dendritic cells.
- Fibroblast and epithelial cells can also take part in efferocytosis.
- Dysregulated efferocytosis is involved in autoimmune diseases and neoplasia

3. Ferroptosis:

- It is a distinct form of cell death that is triggered when excessive intracellular levels of iron or reactive oxygen species cause unchecked membrane lipid peroxidation due to glutathione-dependant antioxidant defenses
- It can be prevented by reducing iron levels
- It results in loss of plasma membrane permeability which ultimately leads to cell death resembling necrosis.
- Microscopically, the most prominent features are the loss of mitochondrial cristae and ruptured outer mitochondrial membrane
- It is linked to cell death in a variety of human pathologies including cancer, neurodegenerative diseases, and stroke.

4. Proapoptotic factors:

- **09th edition:** Has all the four BH domains (BH 1-4)
- **10th edition:** Has only the first three BH domains (BH 1-3)

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